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DRYER FILTER ASSEMBLY AND METHODS FOR REMOVING LINT FROM A DRYER FILTER ASSEMBLY

Abstract

A dryer filter assembly includes a body having a distal end and a proximal end and further defining a frame that receives a filter. The dryer filter assembly further includes a plurality of receptacles, with each of the distal end of the body and the proximal end of the body defining at least one of the plurality of receptacles. A plurality of containers are releasably received by the plurality of receptacles, and a sweeper mechanism is slidably coupled to the body, such that the sweeper mechanism may translate in a longitudinal direction between the distal end and the proximal end of the body. Translation of the sweeper mechanism between the distal end and the proximal end of the body forces particulate accumulated on the filter to be disposed in at least one of the plurality of containers.

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Background/Summary

TECHNICAL FIELD

[0001] The present specification generally relates to household appliances, and more specifically, to lint filters for clothing dryers and methods of removing lint from filters.

BACKGROUND

[0002] In conventional clothing dryers, lint collection is typically managed through a simple, removable lint screen or filter located near an air outlet of the dryer. In these dryers, users must manually clean the filter after one and/or several drying cycles to prevent lint buildup, which can impair dryer performance, increase energy consumption, and pose significant fire risks due to the flammability of lint. Furthermore, these traditional dryers do not provide users with a way to determine when the lint screen and/or filter is full. Accordingly, a need exists for a dryer filter assembly that enhances user convenience while improving dryer performance and safety.

SUMMARY

[0003] In an embodiment, a dryer filter assembly is disclosed. The dryer filter assembly includes a body having a distal end and a proximal end and further defining a frame that receives a filter. The dryer filter assembly further includes a plurality of receptacles, with each of the distal end of the body and the proximal end of the body defining at least one of the plurality of receptacles. A plurality of containers are releasably received by the plurality of receptacles, and a sweeper mechanism is slidably coupled to the body, such that the sweeper mechanism may translate in a longitudinal direction between the distal end and the proximal end of the body. Translation of the sweeper mechanism between the distal end and the proximal end of the body forces particulate accumulated on the filter to be disposed in at least one of the plurality of containers.

[0004] In another embodiment, a dryer filter assembly is disclosed. The dryer filter assembly includes a body having a distal end and a proximal end and further defining a frame configured to receive a filter, a plurality of receptacles, with each of the distal end of the body and the proximal end of the body defining at least one of the plurality of receptacles, at least one receptacle opening formed in each of the plurality of receptacles, a plurality of containers releasably received by the plurality of receptacles, at least one container aperture formed in each of the plurality of containers, such that the at least one container aperture and the at least one receptacle opening are aligned when the plurality of containers are positioned within the plurality of receptacles, and a sweeper mechanism including a sweeper component and a translation component, the sweeper mechanism being slidably coupled to the body such that the sweeper mechanism may translate in a longitudinal direction between the distal end and the proximal end of the body. When the sweeper mechanism translates between the distal end and the proximal end of the body, the sweeper component of the sweeper mechanism contacts a surface of the filter to dislodge lint from the surface of the filter and force the lint into at least one of the plurality of containers.

[0005] In yet another embodiment, a method of removing lint from a dryer filter assembly is disclosed. The method involves positioning at least first container in a first receptacle formed on a proximal end of a body of the dryer filter assembly and a second container a second receptacle formed on a distal end of the body of the dryer filter assembly; translating a sweeper mechanism slidably engaged to a frame of the body of the dryer filter assembly between a proximal end and a distal end of the body, such that the sweeper mechanism contacts a surface of a filter disposed within the frame; dislodging lint accumulated on the surface of the filter with the sweeper mechanism; forcing, via translation of the sweeper mechanism, the dislodged lint into at least one of the first container or the second container; removing at least one of the first container or the second container from the first receptacle or the second receptacle; and disposing of the dislodged lint captured in at least one of the first container or the second container.

[0006] These and additional features provided by the embodiments described herein will be more fully understood in view of the following detailed description, in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The embodiments set forth in the drawings are illustrative and exemplary in nature and not intended to limit the subject matter defined by the claims. The following detailed description of the illustrative embodiments can be understood when read in conjunction with the following drawings, where like structure is indicated with like reference numerals and in which:

[0008] FIG. 1 depicts a front-side view of a dryer filter assembly, according to one or more embodiments shown and described herein;

[0009] FIG. 2A depicts a perspective view of the dryer filter assembly of FIG. 1 with a sweeper mechanism in an initial position, according to one or more embodiments shown and described herein;

[0010] FIG. 2B depicts a perspective view of the dryer filter assembly of FIG. 2A with the sweeper mechanism translating during a lint removal process, according to one or more embodiments shown and described herein;

[0011] FIG. 3 depicts a partially exploded view of the dryer filter assembly of FIG. 1 including a lint collector detached from the dryer filter assembly, according to one or more embodiments shown and described herein;

[0012] FIG. 4 depicts a partially exploded view of another lint collector of the dryer filter assembly of FIG. 1, according to one or more embodiments shown and described herein;

[0013] FIG. 5 depicts a partially exploded view of another lint collector of the dryer assembly of FIG. 1, according to one or more embodiments shown and described herein;

[0014] FIG. 6 depicts a perspective view of another embodiment of the dryer filter assembly of FIG. 1, according to one or more embodiments shown and described herein; and

[0015] FIG. 7 depicts an illustrative flow diagram of a method of removing lint from the dryer filter assembly of FIG. 1, according to one or more embodiments shown and described herein.

DETAILED DESCRIPTION

[0016] Embodiments disclosed herein relate to dryer filter assemblies and methods of removing lint from a dryer filter assembly. In these embodiments, the dryer filter assembly includes a body having a distal end and a proximal end and further defining a frame that receives a filter. The dryer filter assembly further includes a plurality of receptacles, with each of the distal end of the body and the proximal end of the body defining at least one of the plurality of receptacles. A plurality of containers are releasably received by the plurality of receptacles, and a sweeper mechanism is slidably coupled to the body, such that the sweeper mechanism may translate in a longitudinal direction between the distal end and the proximal end of the body. Translation of the sweeper mechanism between the distal end and the proximal end of the body forces particulate accumulated on the filter to be disposed in at least one of the plurality of containers. In these embodiments, the plurality of containers may allow for easy and hygienic disposal of lint, while the sweeper mechanism may ensure efficient lint removal and filter maintenance of the dryer filter assembly.

[0017] As should be appreciated, in traditional clothing dryers, lint collection is managed using a simple, removable lint screen or filter that may be manually cleaned by a user following one and/or several drying cycles in order to prevent lint buildup and mitigate fire risk due to the flammability of lint. However, the frequent manual cleaning required to maintain these traditional lint screens/filters is inconvenient and often overlooked by users, leading to increased fire risk and decreased dryer efficiency. Furthermore, these traditional lint screens/filters may be difficult to access, and may offer inconsistent lint capture during operation of the clothing dryer, which may

result in the buildup of residual lint throughout the filter and/or clothing dryer and the formation of clogs.

[0018] The disclosed dryer filter assembly aims to resolve these issues by including a number of integrated lint containers formed within a body of the dryer filter assembly, which may be removed from the body to allow for easy and hygienic removal of lint from the dryer filter assembly. In these embodiments, the dryer filter assembly further includes a sweeper mechanism, which may be translated across a filter disposed within the body of the dryer filter assembly to transfer lint from the filter and into the lint containers for removal.

[0019] Embodiments of dryer filter assemblies and methods of removing lint from dryer filter assemblies will now be described in additional detail herein. The following will now describe these refrigeration systems, container assemblies, and methods in more detail with reference to the drawings and where like numbers refer to like structures.

[0020] As depicted in FIGS. 1-2B, a dryer filter assembly **10** is depicted. In these embodiments, the dryer filter assembly **10** may include a body **20** having at least a proximal end **22** and a distal end **24**, with the body **20** defining a frame **30** for receiving a filter, such as a dryer lint filter. As further illustrated in FIG. 1, each of the proximal end **22** and the distal end **24** of the dryer filter assembly **10** may further define a receptacle **40**, which may be configured to receive at least one of a plurality of containers **50**, such as a lint containers, as will be described in additional detail herein.

[0021] In the embodiments described herein, it should be appreciated that the body **20** may be formed of any durable and/or lightweight material that may withstand the high temperatures and/or humidity levels typical of a clothing dryer environment. For example, the body **20** may be formed of high temperature plastics (e.g., polycarbonate, polphenylene sulfide, polyetherimide, etc.), metal alloys (e.g., stainless steel, aluminum alloys, etc.), silicone, composite materials (e.g., fiberglass-reinforced plastics, carbon fiber composites, etc.), thermoset plastics, and/or any other similar material without departing from the scope of the present disclosure.

[0022] Referring still to FIGS. 1-2B, the dryer filter assembly **10** may further include a sweeper mechanism **60** that may be slidably coupled to the body **20** of the dryer filter assembly **10**, such that the sweeper mechanism **60** may translate in a longitudinal direction (e.g., in the $\pm$ x-direction as depicted in the coordinate axis of FIG. 1) between the proximal end **22** and the distal end **24** of the body. In these embodiments, as the sweeper mechanism **60** is translated across a length of the body **20** of the dryer filter assembly **10** (e.g., between the proximal end **22** and the distal end **24**), the sweeper mechanism **60** may engage a surface of the filter (not depicted) disposed within the frame **30** of the body **20**, such that contact between the sweeper mechanism **60** and the lint accumulated on the surface of the filter causes the lint to be pushed into at least one of the plurality of containers **50**, as will be described in additional detail herein with reference to FIGS. 2A and 2B.

[0023] In the embodiments described herein, the body **20** may further include a partition **26**, such as a central partition, which may divide the frame **30** into a first frame portion **32** and a second frame portion **34**. In these embodiments, the partition **26** may provide structural rigidity and/or support to the frame **30** of the body **20**, which may aid in preventing flexing and/or warping of the frame **30** that may be otherwise caused by heat within the clothing dryer and/or during removal and reinsertion of the dryer filter assembly **10**. Furthermore, in some embodiments, the partition **26** may aid in evenly distributing lint across the surface of the filter, which may ensure that lint is evenly dispersed in the plurality of containers **50**, as will be described in additional detail herein with reference to FIGS. 2A and 2B.

[0024] Referring still to FIGS. 1-2B, the dryer filter assembly **10** may further include a cover **70**, which may be used to enclose the frame **30** and secure a filter within the frame **30**. In these embodiments, the cover **70** may be formed as a two-part cover, such that the cover **70** includes a first cover portion **72** and a second cover portion **74**. As illustrated in FIG. 1, the cover **70** may further include a latch **76**, such as a cover latch, that may be configured to releasably and/or detachably couple the first cover portion **72** to the second cover portion **74**. In these embodiments,

releasing the latch **76** may allow for the first cover portion **72** to be disengaged from the second cover portion **74**, such that a user may access the frame **30** of the dryer filter assembly **10** to replace a filter without needing to remove the dryer filter assembly **10** from the dryer.

[0025] As further depicted in FIGS. **1-2B**, the cover **70** may further include a plurality of perforations **78**. In these embodiments, the plurality of perforations **78** may allow for air to flow through the dryer filter assembly **10**. For example, during operation of a clothing dryer, air may pass through the dryer filter assembly **10** to exit the clothing dryer. Accordingly, the plurality of perforations **78** may ensure that air may pass freely through the dryer filter assembly **10** during operation of the clothing dryer.

[0026] Furthermore, in the embodiments described herein, the plurality of perforations **78** may be designed to maximize a volume of lint that may be trapped within the dryer filter assembly **10** while minimizing the obstruction of air passage through the dryer filter assembly **10**. For example, the plurality of perforations **78** may be sized and positioned such that the plurality of perforations **78** are small enough to capture lint within the dryer filter assembly **10** while being large enough to allow for the passage of moist air through the dryer filter assembly **10** during operation of the clothing dryer.

[0027] Referring still to FIGS. **1-2B**, it should be further appreciated the size and/or positioning of the plurality of perforations **78** on the cover **70** of the dryer filter assembly **10** may aid in preventing excessive lint accumulation within the frame **30** of the dryer filter assembly **10**. For example, in these embodiments, the plurality of perforations **78** may further act to limit excessive lint accumulation in any particular portion of the dryer filter assembly **10**, thereby allowing for even lint distribution across the filter positioned within the frame **30** of the dryer filter assembly **10** and minimizing the risk of clogging in the clothing dryer.

[0028] Referring now to FIGS. **2A** and **2B**, operation of the dryer filter assembly **10** will be described in additional detail. Initially, as a drum of the clothing dryer rotates, articles (e.g., clothing or otherwise) within the clothing dryer may begin to tumble, thereby causing friction between the articles and leading to the formation of lint (e.g., via fibers being loosened and/or shed from the articles). As the clothing dryer continues its operation cycle, air is pulled into the clothing dryer from an external environment, heated, and passed through the clothing dryer. As the heated air flow passes through the clothing dryer, it may also carry lint generated by the contact between articles during the tumble process.

[0029] In these embodiments, the heated air carrying the lint is drawn through the dryer filter assembly **10** as it exits the clothing dryer. Accordingly, as the heated air passes through the dryer filter assembly **10**, lint carried by the heated air is captured by the filter positioned within the frame **30** of the body **20** of the dryer filter assembly **10**. Over the course of an operation cycle, lint may be accumulated on the filter of the dryer filter assembly **10**, as has been described herein.

[0030] Depending on the amount of use and the types of articles being dried, the accumulation of lint on the filter over a number of operational cycles of the clothing dryer may begin to reduce the permeability and effectiveness of the dryer filter assembly **10**. At this point, it may be necessary to clean the filter (e.g., dispose of the lint accumulated on the filter) in order to regain desired levels of permeability and efficiency.

[0031] Referring still to FIGS. **2A** and **2B**, the lint accumulated on the filter may be disposed into at least one of the plurality of containers **50** by translating the sweeper mechanism **60** across a surface of the filter, such that contact between the sweeper mechanism **60** and the filter surface causes accumulated lint to be pushed into at least one of the plurality of containers **50**. In these embodiments, the frame **30** of the body **20** of the dryer filter assembly **10** may include a slot **36**, which may be used to guide translation of the sweeper mechanism **60** across the surface of the filter, as will be described in additional detail herein.

[0032] In the embodiments described herein, the sweeper mechanism **60** may further include a sweeper component **62** and a translation component **64**. As illustrated most clearly in FIGS. **2A** and

2B, the sweeper component **62** of the sweeper mechanism **60** may be the portion of the sweeper mechanism **60** that contacts the surface of the filter, while the translation component **64** may be manipulated to translate the sweeper component **62** in the longitudinal direction (e.g., in the +/- direction as depicted in the coordinate axes of FIGS. 2A and 2B) across the surface of the filter. In the embodiments described herein, it should be appreciated that the sweeper component **62** of the sweeper mechanism **60** may include a rubber edge, a felted rubber edge, a brush (e.g., a brush including a plurality of bristles, etc.), or any other similar sweeper component that is configured to contact a surface of the filter and dislodge lint accumulated on the filter, such that the sweeper component **60** may be used to direct lint into the plurality of containers **50**, as will be described herein.

[0033] Referring still to FIGS. 2A and 2B, and in the embodiments described herein, the translation component **64** of the sweeper mechanism **60** may be manually translated through the slot **36** defined in the frame **30** of the dryer filter assembly **10** such that the sweeper component **62** traverses the surface of the filter and disposes of accumulated lint. In these embodiments, the translation component **64** may be manually translatable (e.g., by grasping of a user, etc.) such that a user may be capable of pushing and/or pulling the translation component **64** to translate the sweeper mechanism **60** across the surface of the filter. Accordingly, after an operational cycle of the clothing dryer, a user may simply grasp the translation component **64** of the sweeper mechanism **60** and move the translation component **64** from the proximal end **22** of the body **20** to the distal end **24** of the body **20** (or vice versa) to dispose of lint accumulated on the filter between operational cycles of the clothing dryer.

[0034] Although not depicted, in some embodiments, the sweeper mechanism **60** may include a controller that may allow for the sweeper mechanism **60** to be automatically translated following each operational cycle of the clothing dryer. For example, in these embodiments, the translation of sweeper component **62** may be driven by an actuator, a motor, or any other similar electromechanical component, such that the sweeper mechanism **60** may be used to dispose of lint accumulated on the filter of the dryer filter assembly **10** without intervention from a user.

[0035] Referring still to FIGS. 2A and 2B, and in the embodiments described herein, the sweeper component **62** of the sweeper mechanism **60** may include a length (e.g., in the +/- y-direction (e.g., in the +/- y-direction as depicted in the coordinate axes of FIGS. 2A and 2B) that is equivalent to a height (e.g., in the +/- y-direction as depicted in the coordinate axes of FIGS. 2A and 2B) of the frame **30** of the dryer filter assembly **10**. In these embodiments, by utilizing a sweeper component **62** with a length that is equivalent to a height of the frame **30**, it may be possible to ensure that the sweeper component **62** of the sweeper mechanism **60** disposes of lint across a full surface area of the filter when the sweeper mechanism **60** is translated, as has been described herein.

[0036] For example, FIG. 2A depicts the sweeper mechanism **60** in an initial position, in which the sweeper mechanism **60** is adjacent the proximal end **22** of the body **20** of the dryer filter assembly **10**. In these embodiments, once the filter disposed within the frame **30** of the dryer filter assembly **10** has accumulated a volume of lint, the sweeper mechanism **60** may be translated (e.g., in the + x-direction as depicted in the coordinate axes of FIGS. 2A and 2B) towards the distal end **24** of the body **20** of the dryer filter assembly **10**, as is depicted in FIG. 2B. It should be appreciated that, as the sweeper mechanism **60** translates from the proximal end **22** to the distal end **24** of the body **20**, lint that is accumulated on the filter may be similarly pushed towards the distal end **24** of the body **20**.

[0037] In these embodiments, the plurality of containers **50** may include a first container **50a** positioned within a first receptacle **40a** formed at the proximal end **22** of the body **20** and a second container **50b** positioned within a second receptacle **40b** formed at the distal end **24** of the body **20**. Accordingly, as the sweeper mechanism **60** translates from the initial position (e.g., adjacent the proximal end **22** of the body **20**) towards the distal end **24** of the body **20**, the sweeper mechanism **60** may force lint accumulated on the filter into the second container **50b** positioned within the

second receptacle **40b**, as will be described in additional detail herein with reference to FIG. 3. [0038] Referring again to FIGS. 2A and 2B, once the sweeper mechanism **60** has translated to the distal end **24** of the body **20** and pushed any accumulated lint into the second container **50b**, the sweeper mechanism **60** may remain adjacent the distal end **24** of the body **20** (e.g., in a second position or translated position) until the filter positioned within the dryer filter assembly **10** has again accumulated a volume of lint that requires removal from the filter. At this point, the sweeper mechanism **60** may be translated from the distal end **24** of the body **20** towards the proximal end **22** of the body (e.g., in the - x-direction as depicted in the coordinate axes of FIGS. 2A and 2B), such that lint accumulated on the filter is forced into the first container **50a** and the sweeper mechanism returns to the initial position.

[0039] Although FIGS. 2A and 2B depict the initial position of the sweeper mechanism **60** as being adjacent the proximal end **22** of the body **20**, it should be appreciated that the sweeper mechanism **60** may include any number of initial positions without departing from the scope of the present disclosure. For example, in some embodiments, the initial position of the sweeper mechanism **60** may be adjacent the distal end **24** of the body **20**, such that the sweeper mechanism **60** is translated from the distal end **24** of the body **20** to the proximal end **22** of the body **20** (e.g. in the - x-direction as depicted in the coordinate axes of FIGS. 2A and 2B) when the sweeper mechanism **60** is moved from the initial position to the translated position. Furthermore, in other embodiments, the initial position of the sweeper mechanism **60** may be aligned with the partition **26** of the body **20**. In these embodiments, the sweeper mechanism **60** may be translated such that lint which is accumulated on a portion of the filter corresponding to the first frame portion **32** of the frame **30** is disposed of in the first container **50a**, while lint accumulated on a portion of the filter corresponding with the second frame portion **34** is disposed of in the second container **50b**. For example, in these embodiments, the sweeper mechanism **60** may be translated from a central position (e.g., in alignment with partition **26**) towards the distal end **24** of the body **20**, such that lint accumulated in the second frame portion **34** is forced into the second container **50b**. The sweeper mechanism **60** may then be translated from the distal end **24** of the body **20** towards the proximal end **22** of the body, such that lint accumulated in the first frame portion **32** is forced into the first container **50a**. At this point, the sweeper mechanism **60** may then be returned to the initial position (e.g., in alignment with the partition **26**).

[0040] In the embodiments described herein, it should be further appreciated that the sweeper component **62** of the sweeper mechanism **60** may be formed of any durable, flexible material that is capable of withstanding the heat and humidity of a typical clothing dryer environment while dislodging lint from a filter. For example, the sweeper component **62** of the sweeper mechanism **60** may be formed of nylon bristles, polyester bristles, silicone, thermoplastic elastomers, polytetrafluoroethylene, natural fibers, metal wires, or any other similar material without departing from the scope of the present disclosure. It should be further noted that, as described herein, the material of the sweeper component **62** may be further determined based on operational requirements of the dryer filter assembly **10** and/or the filter disposed within the frame **30** of the dryer filter assembly **10**.

[0041] Referring now to FIG. 3, the plurality of containers **50** of the dryer filter assembly **10** will be described in additional detail. As noted herein, the plurality of containers **50** may be releasably received within the receptacles **40** formed at the proximal end **22** and the distal end **24** of the body **20**. For example, and as described herein, the plurality of containers **50** may include the first container **50a** received by the first receptacle **40a** and the second container **50b** received by the second receptacle **40b**.

[0042] In the embodiments described herein, the receptacles **40** may be sized and/or shaped to match an exterior profile of the plurality of containers **50**. For example, as depicted in FIG. 3, the plurality of containers **50** and the receptacles **40** may each have a generally tubular and/or cylindrical profile, such that the plurality of containers **50** may fit snugly within the receptacles **40**

while being easily removable to dispose of lint accumulated within the plurality of containers **50**. Although not depicted, in some embodiments, the receptacles **40** may further include guiding components, such as grooves and/or rails, that are configured to receive the plurality of containers **50** and ensure alignment of particular components of the plurality of containers **50** and the receptacles, as will be described in additional detail herein.

[0043] Referring still to FIG. 3, in order to allow for lint accumulated on the filter to be pushed into the plurality of containers **50** via the sweeper mechanism **60**, as has been described herein, the receptacles **40** may each include a receptacle opening **42** that may be aligned with a container aperture **52** formed in each of the plurality of containers **50** when the plurality of containers **50** are positioned within the receptacles **40**. For example, as depicted in FIG. 3, the first receptacle **40a** may include a first receptacle opening **42a**, which may be in alignment with a first container aperture **52a** formed in the first container **50a**. Accordingly, when the sweeper mechanism **60** translates from the distal end **24** of the body **20** towards the proximal end **22** of the body **20** (e.g., in the-x-direction as depicted in the coordinate axis of FIG. 3), the lint accumulated on the filter may pass through the first receptacle opening **42a** of the first receptacle **40a** and the first container aperture **52a** of the first container **50a** and be stored within the first container **50a** prior to disposal. Similarly, the second receptacle **40b** may include a second receptacle opening **42b** which may be alignment with a second container aperture **52b** formed in the second container **50b** when the second container **50b** is positioned within the second receptacle **40b**. Accordingly, when the sweeper mechanism **60** translates from the proximal end **22** of the body **20** towards the distal end **24** of the body **20** (e.g., in the +x-direction as depicted in the coordinate axis of FIG. 3), lint accumulated on the filter may pass through the second receptacle opening **42b** of the second receptacle **40b** and the second container aperture **52b** of the second container **50b** and be stored within the second container **50b** prior to disposal.

[0044] Although the plurality of containers **50** and the receptacles **40** depicted in FIG. 3 are illustrated as being cylindrical in shape, it should be appreciated that the plurality of containers **50** may take any shape without departing from the scope of the present disclosure. For example, in embodiments, the plurality of containers **50** may be rectangular, triangular, or any other similar shape without departing from the scope of the present disclosure. Furthermore, it should be understood that the receptacles **40** may take any shape that corresponds to the shape of the container of the plurality of containers **50** that is received by the receptacle **40**. Additionally, it should be noted that, in some embodiments, each of the plurality of containers **50** may take different shapes. For example, the first container **50a** may be cylindrical while the second container **50b** may be rectangular. Accordingly, in these embodiments, the receptacles **40** may similarly include different shapes that correspond to the shapes of the container of the plurality of containers **50** that is received by the receptacle **40**.

[0045] Referring still to FIG. 3, in these embodiments, the plurality of containers **50** may each include a cap **54** that may be releasably coupled a respective container of the plurality of containers **50**. For example, as illustrated in FIG. 3, the first container **50a** may include a first cap **54a** while the second container **50b** may include a second cap **54b**. In these embodiments, the cap **54** may ensure that lint which is disposed within each of the plurality of containers **50** does not escape from the plurality of containers **50** until the plurality of containers **50** are removed from the dryer filter assembly **10** to dispose of the lint. In these embodiments, the cap **54** may be coupled to the plurality of containers **50** via any suitable releasably coupling, including a snap-fit coupling, threaded coupling, or any other similar coupling. When the plurality of containers **50** are removed from the dryer filter assembly **10**, the cap **54** of each respective container of the plurality of containers **50** may be removed, such that lint accumulated within each of the plurality of containers **50** may be disposed.

[0046] Referring still to FIG. 3, in some embodiments, the dryer filter assembly **10** may further include a plurality of sensor mechanisms **80** that may be configured to monitor the accumulation of

lint within the dryer filter assembly **10** and notify a user that lint within the dryer filter assembly **10** requires disposal.

[0047] In these embodiments, and as depicted in FIG. 3, the plurality of sensor mechanisms **80** may be disposed on each of the plurality of containers **50**, and may be configured to monitor a volume of lint accumulated within each of the plurality of containers **50**. For example, a first sensor mechanism **80a** may be integrated into and/or otherwise disposed on the first container **50a**, while a second sensor mechanism **80b** may be integrated into and/or otherwise disposed on the second container **50b**. In these embodiments, the first sensor mechanism **80a** may monitor a first volume of lint contained in the first container **50a**, while the second sensor mechanism **80b** may monitor a second volume of lint contained in the second container **50b**. In these embodiments, when the first sensor mechanism **80a** determines that the first volume of lint contained in the first container **50a** exceeds a threshold lint volume, or when the second sensor mechanism **80b** determines that the second volume of lint contained in the second container **50b** exceeds the threshold lint volume, the first and/or second sensor mechanism **80a**, **80b** may provide a notification to a user indicating that the first and/or second container **50a/50b** require cleaning. In the embodiments described herein, the notification provided to the user may be in the form of an audible tone generated by the clothing dryer, or, in more sophisticated and/or smart clothing dryer systems, may include a mobile and/or computer notification provided to a mobile and/or computer application associated with the clothing dryer and/or dryer filter assembly **10**.

[0048] Referring still to FIG. 3, in some embodiments, the plurality of sensors **80** may be further communicatively coupled to the sweeper mechanism in order to monitor (e.g., count) a number of times the sweeper mechanism **60** translates between the proximal end **22** and the distal end **24** of the body **20**. In these embodiments, the plurality of sensors **80** may generate a notification to a user indicating that the plurality of containers **50** require emptying by counting a number of translations of the sweeper mechanism **60** rather than by monitoring a volume of lint within each of the plurality of containers **50**. In the embodiments described herein, when the number of translations meets and/or exceeds a translation threshold, the plurality of sensors **80** may generate the notification to the user, as has been described in detail herein.

[0049] Referring now to FIGS. 4 and 5, in some embodiments, the plurality of containers **50** may include additional features that aid in retention and/or removal of lint disposed in each of the plurality of containers **50**. For example, as illustrated in FIG. 4, the plurality of containers **50** may each include a canister **56**. In these embodiments, each of the plurality of containers **50** may receive a canister **56** which occupies an internal volume of each of the plurality of containers **50**, such that lint which is forced into each of the plurality of containers **50** is received within each respective canister **56**.

[0050] Referring still to FIG. 4, in these embodiments, the canister **56** may be rotatably received within each of the plurality of containers **50**. For example, the canister **56** may be sized and/or shaped such that, when the canister **56** is positioned in any one of the plurality of containers **50**, the canister **56** may be capable of rotating about its lateral axis (e.g., the y-axis as depicted in FIG. 4).

[0051] As further depicted in FIG. 4, the canister **56** may include a canister aperture **58**, which may be rotated into and out of alignment with the container aperture **52** of the container in which the canister **56** is positioned. Accordingly, rotating the canister **56** such that the container aperture **56** is aligned with the canister aperture **58** may allow and/or prevent lint from entering and/or exiting each of the plurality of containers **50**, as will be described in additional detail herein.

[0052] For example, the canister **56** may be rotated such that the canister aperture **58** is in alignment with the container aperture **52** of the container in which the canister **56** is positioned. In these embodiments, with the canister aperture **58** and the container aperture **52** aligned, lint may be forced into the canister **56** using the sweeper mechanism **60**, as has been described in detail herein with reference to FIGS. 1 and 2. However, once the canister **56** is filled with lint, the canister **56** may be rotated, such that the canister aperture **58** is misaligned with the container aperture **52**,

thereby preventing lint from entering the canister **56**. It should be appreciated that, in these embodiments, rotating the canister **56** prior to removal of each of the plurality of containers **50** from the dryer filter assembly **10** may aid in ensuring that lint does not escape from the plurality of containers **50** during the removal process.

[0053] Referring still to FIG. **4**, it should be further understood that, in these embodiments, the canister **56** may be coupled to and/or integrally formed with the cap **54** of each of the plurality of containers **50**, such that rotation of the cap **54** similarly results in rotation of the canister **56**, as has been described herein.

[0054] Referring now to FIG. **5**, in these embodiments, each of the plurality of containers **50** may further include a plunger **90**, which may be configured to aid in removing lint from each of the plurality of containers **50** once the plurality of containers **50** are removed from the dryer filter assembly **10**.

[0055] In these embodiments, the plunger **90** may include a first end **92** and a second end **94**, with the first end **92** being coupled to at least a portion of the cap **54** and the second end **94** including a plunger plate **96**. As depicted in FIG. **5**, the plunger **90** may be inserted into each of the plurality of containers **50** such that the plunger plate **96** contacts a bottom surface (e.g., floor, etc.) of the container of the plurality of containers **50** in which the plunger **90** is disposed. In these embodiments, the plunger **90** may be translatable along a lateral axis (e.g., the y-axis as depicted in FIG. **5**), such that the plunger plate **96** is capable of moving in the lateral direction along a height of the container in which the plunger plate **96** is disposed.

[0056] As should be further appreciated in view of FIG. **5**, lint may be received by each of the plurality of containers **50** using the sweeper mechanism **60**, as has been described in detail herein with reference to FIGS. **1** and **2A-2B**. In these embodiments, when lint from the filter is forced into the container of the plurality of containers **50** that includes the plunger **90**, the lint may collect above the plunger plate **96**. Once the container is sufficiently full (e.g., as may be determined by the plurality of sensor mechanisms **80**), the container may be removed from the dryer filter assembly **10**. To remove the lint from the container, the cap **54** may be pulled (e.g., in the + y-direction as depicted in the coordinate axis of FIG. **5**), which may in turn cause the plunger plate **96** to traverse the container. As the plunger plate **96** moves, the plunger plate **96** may contact lint disposed above the plunger plate **96**, such that the plunger plate **96** forces lint within the container to be forced out of the opening formed in the top portion of the container via the removal of the cap **54**.

[0057] Referring now to FIG. **6**, another embodiment of a dryer filter assembly **10** is depicted. In these embodiments, the body **20** of the dryer filter assembly **10** may be formed of at least two parts, such as a first body member **20a** and a second body member **20b**, with the first body member **20a** and the second body member **20b** being releasably coupled. For example, in these embodiments, the body **20** may include a hinge **28** disposed on a bottom surface of the body **20** which allows for the first body member **20a** and the second body member **20b** to be moved from a closed position (e.g., in which the latch **76** of the cover **70** secures the first body member **20a** and the second body member **20b** together) and an open position (e.g., in which the latch **76** is released such that the first body member **20a** and the second body member **20b** translate relative one another).

[0058] In these embodiments, the plurality of containers **50** may be fixedly attached and/or integrally coupled to the proximal end **22** and the distal end **24** of the body **20**, such that the plurality of containers **50** may not be removed from the body **20**. Accordingly, in these embodiments, in order to remove lint that has been accumulated within each of the plurality of containers **50** from the dryer filter assembly **10**, the first and second body members **20a**, **20b** may be translated from the closed position to the open position, such that the lint within the plurality of containers **50** may be accessed and removed.

[0059] Although FIG. **6** depicts the body **20** of the dryer filter assembly **10** as including a hinge **28**, it should be appreciated that the body **20** may include any coupling (e.g., snap fit, etc.) that may

allow the first body member **20a** and the second body members **20b** to move relative one another without departing from the scope of the present disclosure.

[0060] Turning now to FIG. 7, an illustrative method **700** of removing lint from a dryer filter assembly is described in additional detail. In these embodiments, the method may initially involve positioning at least a first container in a first receptacle formed on a proximal end of a body of the dryer filter assembly and a second container in a second receptacle formed on a distal end of the body of the dryer filter assembly, as is illustrated at block **710**.

[0061] With the plurality of containers (e.g., first and/or second container) positioned within their respective receptacles, the method may advance to block **720**, which may involve translating a sweeper mechanism between a proximal end and a distal end of the body, such that the sweeper mechanism contacts a surface of a filter disposed within a frame of the body. In these embodiments, the translation of the sweeper mechanism may further result in dislodging of lint accumulated on the surface of the filter, as depicted at block **730**.

[0062] As lint is dislodged from the filter and the sweeper mechanism continues to translate, the method may proceed to block **740**, which may involve forcing (e.g., via translation of sweeper mechanism) the dislodged lint into at least one of the plurality of containers (e.g., the first and/or second container). In these embodiments, once the lint is dislodged from the filter and forced into at least one of the containers, the method may proceed to block **750**, which may involve removing at least one of the first or second containers from the first or second receptacle, as may be dictated based on the container that received the dislodged lint. After the appropriate container has been removed from the dryer filter assembly, the method may advance to block **760**, which may involve disposing of the dislodged lint captured in the container.

[0063] As should be appreciated in view of the foregoing, a dryer filter assembly is disclosed. The dryer filter assembly includes a body having a distal end and a proximal end and further defining a frame that receives a filter. The dryer filter assembly further includes a plurality of receptacles, with each of the distal end of the body and the proximal end of the body defining at least one of the plurality of receptacles. A plurality of containers are releasably received by the plurality of receptacles, and a sweeper mechanism is slidably coupled to the body, such that the sweeper mechanism may translate in a longitudinal direction between the distal end and the proximal end of the body. Translation of the sweeper mechanism between the distal end and the proximal end of the body forces particulate accumulated on the filter to be disposed in at least one of the plurality of containers. In these embodiments, the plurality of containers may allow for easy and hygienic disposal of lint, while the sweeper mechanism may ensure efficient lint removal and filter maintenance of the dryer filter assembly.

[0064] The terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms, including “at least one,” unless the content clearly indicates otherwise. “Or” means “and/or.” As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” or “includes” and/or “including” when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof. The term “or a combination thereof” means a combination including at least one of the foregoing elements.

[0065] It is noted that the terms “substantially” and “about” may be utilized herein to represent the inherent degree of uncertainty that may be attributed to any quantitative comparison, value, measurement, or other representation. These terms are also utilized herein to represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the basic function of the subject matter at issue.

[0066] While particular embodiments have been illustrated and described herein, it should be

understood that various other changes and modifications may be made without departing from the spirit and scope of the claimed subject matter. Moreover, although various aspects of the claimed subject matter have been described herein, such aspects need not be utilized in combination. It is therefore intended that the appended claims cover all such changes and modifications that are within the scope of the claimed subject matter.

Claims

1. A dryer filter assembly comprising: a body having a distal end and a proximal end and further defining a frame configured to receive a filter; a plurality of receptacles, with each of the distal end of the body and the proximal end of the body defining at least one of the plurality of receptacles; a plurality of containers releasably received by the plurality of receptacles; and a sweeper mechanism slidably coupled to the body, such that the sweeper mechanism may translate in a longitudinal direction between the distal end and the proximal end of the body; wherein translation of the sweeper mechanism between the distal end and the proximal end of the body forces particulate accumulated on the filter to be disposed in at least one of the plurality of containers.
2. The dryer filter assembly of claim 1, wherein the sweeper mechanism further includes a translation mechanism for manually translating the sweeper mechanism between the proximal end and the distal end of the body.
3. The dryer filter assembly of claim 1, wherein the body further include a partition that divides the frame into a first frame portion and a second frame portion.
4. The dryer filter assembly of claim 1, wherein the plurality of containers each include a container aperture.
5. The dryer filter assembly of claim 4, wherein the plurality of receptacles each include a receptacle opening.
6. The dryer filter assembly of claim 5, wherein the container aperture of each of the plurality of containers is aligned with the receptacle opening of each of the plurality of receptacles when the plurality of containers are positioned within the plurality of receptacles.
7. The dryer filter assembly of claim 1, wherein each of the plurality of containers further includes a cap releasably coupled to the plurality of containers.
8. The dryer filter assembly of claim 1, wherein the sweeper mechanism further includes a sweeper component configured to contact a surface of the filter when the sweeper mechanism translates between the proximal end and the distal end of the body.
9. The dryer filter assembly of claim 8, wherein the sweeper component has a sweeper component length that is equivalent to a frame height of the frame.
10. The dryer filter assembly of claim 1, further comprising a cover configured to enclose the frame and secure the filter within the frame.
11. The dryer filter assembly of claim 10, wherein the cover includes a first cover portion and a second cover portion, the first cover portion and the second cover portion being releasably coupled via a latch.
12. The dryer filter assembly of claim 10, wherein the cover further includes a plurality of perforations.
13. The dryer filter assembly of claim 1, further including a plurality of sensor mechanisms configured to monitor a volume of lint in each of the plurality of containers.
14. The dryer filter assembly of claim 13, wherein, when the plurality of sensor mechanisms determine that the volume of lint in at least one of the plurality of containers exceeds a threshold lint volume, the plurality of sensor mechanisms generate a notification to a user.
15. The dryer filter assembly of claim 1, further comprising a plurality of sensor mechanisms configured to monitor a number of translations of the sweeper mechanism between the proximal end and the distal end of the body.

16. The dryer filter assembly of claim 15, wherein, when the plurality of sensor mechanisms determine that the number of translations exceeds a translation threshold, the plurality of sensor mechanisms generate a notification to a user.

17. The dryer filter assembly of claim 1, wherein the frame further defines a slot configured to guide translation of the sweeper mechanism between the proximal end and the distal end of the body.

18. A dryer filter assembly comprising: a body having a distal end and a proximal end and further defining a frame configured to receive a filter; a plurality of receptacles, with each of the distal end of the body and the proximal end of the body defining at least one of the plurality of receptacles; at least one receptacle opening formed in each of the plurality of receptacles; a plurality of containers releasably received by the plurality of receptacles; at least one container aperture formed in each of the plurality of containers, such that the at least one container aperture and the at least one receptacle opening are aligned when the plurality of containers are positioned within the plurality of receptacles; and a sweeper mechanism including a sweeper component and a translation component, the sweeper mechanism being slidably coupled to the body such that the sweeper mechanism may translate in a longitudinal direction between the distal end and the proximal end of the body; wherein, when the sweeper mechanism translates between the distal end and the proximal end of the body, the sweeper component of the sweeper mechanism contacts a surface of the filter to dislodge lint from the surface of the filter and force the lint into at least one of the plurality of containers.

19. The dryer filter assembly of claim 18, further including a plurality of sensor mechanisms configured to monitor a volume of lint in each of the plurality of containers.

20. A method of removing lint from a dryer filter assembly comprising: positioning at least a first container in a first receptacle formed on a proximal end of a body of the dryer filter assembly and a second container in a second receptacle formed on a distal end of the body of the dryer filter assembly; translating a sweeper mechanism slidably engaged to a frame of the body of the dryer filter assembly between the proximal end and the distal end of the body, such that the sweeper mechanism contacts a surface of a filter disposed within the frame; dislodging lint accumulated on the surface of the filter with the sweeper mechanism; forcing, via translation of the sweeper mechanism, the dislodged lint into at least one of the first container or the second container; removing at least one of the first container or the second container from the first receptacle or the second receptacle; and disposing of the dislodged lint captured in at least one of the first container or the second container.
